Christopher J Rickerd

231.429.8953
cjrickerd@gmail.com

Professional Experience

Centauri (Formerly Integrity Applications, Inc.) — Ann Arbor, MI
Software Engineer — Nov 2017 - Aug 2020

- Implemented, updated, and tested reference processor infrastructure for synthetic aperture radar image processing

Michigan Tech Research Institute — Ann Arbor, MI
Assistant Research Engineer I — Aug 2016 - Nov 2017

- Applied physics-based modeling techniques and 3D image reconstruction methods for ground-penetrating radar target recognition research

Michigan Tech Research Institute — Ann Arbor, MI
Engineering Intern — Jan 2014 - Aug 2014

- Applied statistical modeling techniques to synthetic aperture radar target recognition research
- Designed drivers, control software, and 3D data visualization software for simultaneous data collection using a camera, automotive radar, and off-the-shelf lidar scanners

Aerophysics, Inc. — Allouez, MI
MISNER Intern — May 2013 - Aug 2013

- Implemented, tested, optimized, and debugged a spacecraft attitude determination system on a DSP board using a Kalman filter to combine data from a star tracker and angular velocity sensors
- Assisted in ion space propulsion lab experiments

Michigan Technological University — Houghton, MI
Aerospace Enterprise Team Member — Jan 2012 - Dec 2013

- Worked as OBDC team member on the University Nanosatellite Program
- Acted in charge of sub-group responsible for developing, testing, and maintaining the satellite's main event loop and command execution code
- Wrote and maintained driver software to interface with on-board flash disk
- Tested hardware-software integration in class-100,000 clean room environment

Software Experience

Languages
C/C++
MATLAB
Java
VB 3.0 / .NET
Shell Script
LabVIEW
Python
Mathematica

IDE's
Eclipse
Dev-C++
Visual Studio
Tornado
PyCharm
IDLE

Operating Systems
UNIX
Windows
VxWorks

Academic Experience

Johns Hopkins University — Baltimore, MD (Online classes)
Completed coursework toward MS Bioinformatics — Summer 2021

University of Maryland — College Park, MD
Completed 2 years of coursework toward Physics PhD — Aug 2014 - May 2016
Passed qualifying exam

Michigan Technological University — Houghton, MI
B.S. Computer Engineering w/ Physics Minor — Aug 2009 - Dec 2013
GPA: 3.96 (Department: 4.0)

Cadillac Senior High School — Cadillac, MI
Class of 2009 — Grad. May 2009

Publications

J.T. Davies, C.J. Rickerd, M.A. Grimes and D.O. Guney, "An n-bit general implementation of Shor's quantum period-finding algorithm," Quantum Information & Computation, 16(7&8), 700–718 (2016). https://arxiv.org/pdf/1411.5949.pdf

B. J. Thelen, C. J. Rickerd, and J. W. Burns, "Novel approach for assessing uncertainty propagation via information-theoretic divergence metrics and multivariate gaussian copula modeling," Algorithms for Synthetic Aperture Radar Imagery, XXI. Vol. 9093, International Society for Optics and Photonics, 2014.